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Test Pattern

ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

NEET(UG)
MINOR
05-02-2023

ENTHUSIAST ADVANCE COURSE PHASE : MEA-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 Hours 20 Minutes** duration and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. In this Test Paper, each subject will consist of **two sections**. **Section A** will consist of **35** questions (all questions are mandatory) and **Section B** will have **15** questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
4. In case of more than one option correct in any question, the best correct option will be considered as answer.
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) ____________________

Form Number : in figures ____________________

: in words ____________________

Centre of Examination (in Capitals) : ____________________

Candidate's Signature : ______________ Invigilator's Signature : ______________

Your Target is to secure Good Rank in Pre-Medical 2024

SUBJECT : PHYSICS

Topic : ELECTROSTATICS, RAY OPTICS

SECTION - A

Attempt All 35 questions

1. If a body is charged by rubbing it, its weight :-

(1) Increases slightly

(2) Decreases slightly

(3) Remain precisely constant

(4) May increase slightly or may decrease slightly

2. When 10^{19} electrons are removed from a neutral metal plate, the electric charge on it is :-

(1) –1.6 C (2) + 1.6 C

(3) 10^{+19} C (4) 10^{-19}C

3. A and B are two conductors, A is positively charged and B is negatively charged. They are connected to the earth by two separate wires. The flow of electrons will be :-

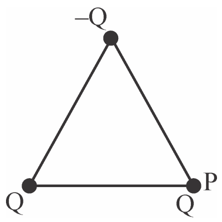
(1) From A towards earth and from earth towards B

(2) From A and B towards the earth

(3) From earth towards A and B

(4) Form earth to A and from B to the earth

4. Find the net force on to the charge placed at point P :

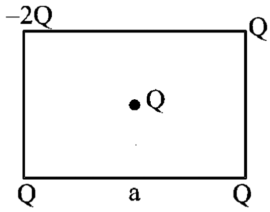


Each side =a

(1) $\frac{kQ^2}{a^2}$ (2) $\frac{\sqrt{3}kQ^2}{a^2}$

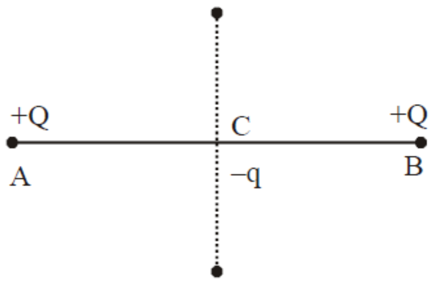
(3) $\frac{2kQ^2}{a^2}$ (4) $\frac{\sqrt{2}kQ^2}{a^2}$

5. Four point charges are placed at different corners of the square of side 'a' as shown, then net force on central charge will be :-



(1) Zero (2) $\frac{6KQ^2}{a^2}$ (3) $\frac{3KQ^2}{a^2}$ (4) $\frac{2KQ^2}{a^2}$

6. Two similar charge of +Q , as shown in figure are placed at A and B. –q charge is placed at point C midway between A and B. –q charge will oscillate if



(1) It is moved towards A.

(2) It is moved towards B.

(3) It is moved upward to AB.

(4) Distance between A and B is reduced.

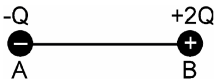
7. Two small conducting spheres of equal radius have charges +10μC and –20μC respectively and placed at a distance R from each other experience force F_1. If they are brough in contact and separated to the same distance, they experience force F_2. The ratio of F_1 to F_2 is:

(1) 1 : 2 (2) –8 : 1 (3) 1 : 8 (4) –2 : 1

8. A charge q is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium, if q is equal to :-

(1) $-Q/2$ (2) $-Q/4$ (3) $+Q/4$ (4) $+Q/2$

9. Charge 2Q and –Q are placed as shown in figure. The point at which electric field intensity is zero will be :



(1) Somewhere between –Q and 2Q

(2) Somewhere on the left of –Q

(3) Somewhere on the right of 2Q

(4) Somewhere on the right bisector of line joining –Q and 2Q

10. A point charge of 100 μC is placed at $3\hat{i}+4\hat{j}$ m. Find the electric field intensity due to this charge at a point located at $9\hat{i}+12\hat{j}$ m.

(1) $8000\ Vm^{-1}$ (2) $9000\ Vm^{-1}$

(3) $2250\ Vm^{-1}$ (4) $4500\ Vm^{-1}$

11. A regular polygon has n sides each of length l. Each corner of the polygon is at a distance r from the centre. Identical charges each equal to q are placed at (n – 1) corners of the polygon. What is the electric field at the centre of the polygon

(1) $\frac{n}{4\pi\varepsilon_0}\frac{q}{r^2}$ (2) $\frac{n}{4\pi\varepsilon_0}\frac{q}{l^2}$

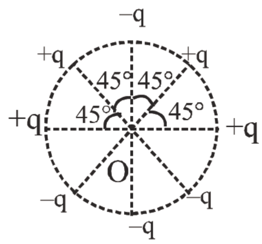
(3) $\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ (4) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$

12. Four charges each Q are placed at four corners of square then what amount of charge should place at center so that all charges comes in equlibrium.

(1) $\frac{-Q}{4}$ (2) $\frac{-Q}{\sqrt{3}}$

(3) $\frac{-Q}{2}\left(\sqrt{2}+\frac{1}{2}\right)$ (4) $\frac{-Q}{4}\left(2+\frac{1}{\sqrt{2}}\right)$

13. Eight charge particles are placed at circumference of circle of radius r. The net electric field at the centre of circle is :-



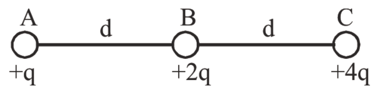
(1) zero (2) $\frac{q}{\sqrt{2}\pi\epsilon_0 r^2}$

(3) $\frac{\sqrt{2}q}{\pi\epsilon_0 r^2}$ (4) $\frac{2q}{\pi\epsilon_0 r^2}$

14. Electrostatic force between two charges is 200 N. When both charges are placed in medium then electrostatic force between them is 80 N. Then ε_r is equal to

(1) 1 (2) 1.25 (3) 2.5 (4) 0

15. Three charges +q, +2q and +4q are connected by strings as shown in the figure. What is the ratio of tensions in the strings AB and BC.



(1) 1 : 2 (2) 1 : 3 (3) 4 :1 (4) 4 : 3

16. A charge particle q_1 is at position (2, –1, 3). The electrostatic force on another charged particle q_2 at (0, 0, 0) is :

(1) $\frac{q_1q_2}{56\pi\epsilon_0}\left(2\hat{i}-\hat{j}+3\hat{k}\right)$

(2) $\frac{q_1q_2}{56\sqrt{14}\pi\epsilon_0}\left(2\hat{i}-\hat{j}+3\hat{k}\right)$

(3) $\frac{q_1q_2}{56\pi\epsilon_0}\left(\hat{j}-2\hat{i}-3\hat{k}\right)$

(4) $\frac{q_1q_2}{56\sqrt{14}\pi\epsilon_0}\left(\hat{j}-2\hat{i}-3\hat{k}\right)$

17. Dielectric constant for metal is

(1) Zero (2) Infinite

(3) 1 (4) Greater than 1

18. Three equal charges are placed on the three corners of a square. If the force between q_1 and q_2 is F_{12} and that between q_1 and q_3 is F_{13}, the ratio of magnitudes $\frac{F_{12}}{F_{13}}$ is :-

(1) 1/2 (2) 2 (3) $1/\sqrt{2}$ (4) $\sqrt{2}$

19. if an observer is walking away from the plane mirror with 6m/sec.Then the speed of the image with respect to observer will be :-

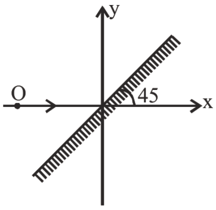
(1) 6 m/sec (2) –6 m/sec

(3) 12 m/sec (4) 3 m/sec

20. Find minimum length of mirror required for a person to see his complete image if person height is 5 feets.

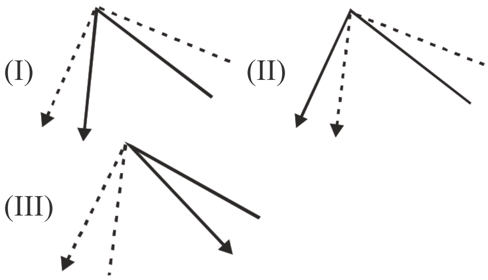
(1) 2 feet (2) 3 feet (3) 4 feet (4) 2.5 feet

21. Image of the object can be :-



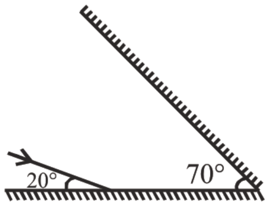
(1) ↓ (2) ↑ (3) → (4) ↖

22. Each of these diagrams is supposed to show two different rays being reflected from the same point on the same plane mirror. Which one of the following is correct ?



(1) Only I (2) Only II (3) Only III (4) All

23. Deviation of emergent ray is :-



(1) 120 (2) 140 (3) 180 (4) 90

24. If a plane mirror is rotated by an angle 8° without changing the direction of incident ray then reflected ray is rotated by :-

(1) 2° (2) 4° (3) 8° (4) 16°

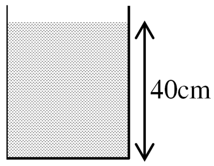
25. A convex mirror has a focal length f. A real object is placed at a distance f in front of it from the pole produces an image at

(1) Infinity (2) f (3) f/2 (4) 2f

26. A square of side 3 cm is placed at a distance of 25 cm from a concave mirror of focal length 10 cm. The centre of the square is at the axis of the mirror and the plane is normal to the axis. The area enclosed by the image of the wire is -

(1) 4 cm^2 (2) 6 cm^2 (3) 16 cm^2 (4) 36 cm^2

27. In the given figure an observer in air (n = 1) sees the bottom of a beaker filled with water (n = 4/3) upto a height of 40 cm. What will be the depth felt by this observer.



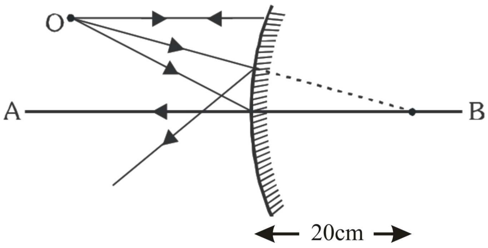
(1) 10 cm (2) 20 cm (3) 30 cm (4) 40 cm

28. The distance of an object from a spherical mirror is equal to the focal length of the mirror then the image :-

(1) must be at infinity (2) may be at infinity

(3) may be at the focus (4) none

29. In the figure a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the diagram? (AB is its principal axis) :-



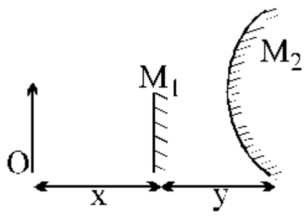
(1) 3 (2) 2 (3) 1 (4) 0

30. A particle is moving with velocity $(\hat{i}+2\hat{j}+3\hat{k})$ m/s. A plane mirror exists in the x-y plane. The mirror has a velocity $(\hat{i}+2\hat{j})$ m/s. Velocity of the image (in m/s) is:-

(1) $(\hat{i}+2\hat{j})$ (2) $\hat{i}+2\hat{j}-3\hat{k}$

(3) $-\hat{j}-\hat{k}$ (4) none of these

31. An object O is placed in front of a small plane mirror M_1 and a large convex mirror M_2 of focal length f. The distance between O and M_1 is x, and the distance between M_1 and M_2 is y. The images of O formed by M_1 and M_2 coincide. The magnitude of f is



(1) $\frac{x^2-y^2}{2y}$ (2) $\frac{x^2+y^2}{2y}$

(3) x - y (4) $\frac{x^2+y^2}{x-y}$

32. A bird in air looks a fish vertically below it and inside water ; h_1 is the height of the bird above the surface of water and h_2 the depth of the fish below the surface of water. If refractive index of water with respect to air be μ, then the distance of the fish as observed by the bird is :

(1) h_1+h_2 (2) $h_1+\frac{h_2}{\mu}$

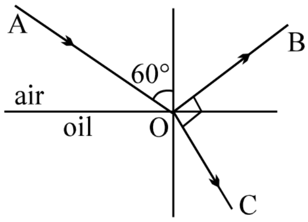
(3) μh_1+h_2 (4) $\mu h_1+\mu h_2$

33. The refractive index of a certain glass is 1.5 for light whose wavelength in vacuum is 6000 Å. The wavelength of this light when it passes through glass is

(1) 4000 Å (2) 6000 Å

(3) 9000 Å (4) 15000 Å

34. A ray of light AO is incident on the surface of oil. Reflected part of this ray OB and refracted part OC are mutually perpendicular as shown. Find refractive index of oil.



(1) $\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{3}}$ (4) 2

35. A ray of light passes from glass having refractive index of 1.6 to air. The angle of incidence for which the angle of refraction is twice the angle of incidence is :-

(1) $\sin^{-1}(4/5)$ (2) $\sin^{-1}(3/5)$

(3) $\sin^{-1}(5/8)$ (4) $\sin^{-1}(2/5)$

SECTION - B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

36. A point charge Q is placed at a point A(x, y) and q at B(0, 0). Find electric force on 'q' in vector form :-

(1) $\dfrac{KQq}{(x^2+y^2)^{3/2}}(x\hat{i}+y\hat{j})$

(2) $\dfrac{KQq}{(x^2+y^2)^{3/2}}(y\hat{i}+x\hat{j})$

(3) $\dfrac{KQq}{(x^2+y^2)^{3/2}}(-x\hat{i}-y\hat{j})$

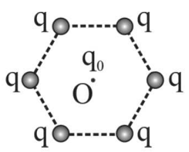
(4) $\dfrac{KQq}{(x^2+y^2)^{3/2}}(-\hat{i}-\hat{j})$

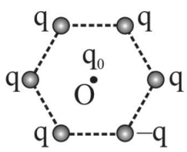
37. Two small spheres each having the charge +Q are suspended by insulating threads of length L from a hook. This arrangement is taken in space where there is no gravitational effect, then the angle between the two suspensions and the tension in each will be :-

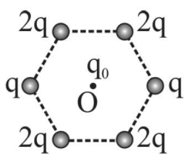
(1) $180^\circ, \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q^2}{(2L)^2}$ (2) $90^\circ, \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q^2}{L^2}$

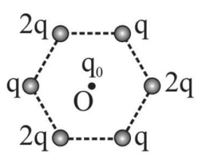
(3) $180^\circ, \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q^2}{2L^2}$ (4) $180^\circ, \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q^2}{L^2}$

38. Figures below show regular hexagons, with charges at the vertices. In which of the following cases the net electric force at the centre is not zero :-

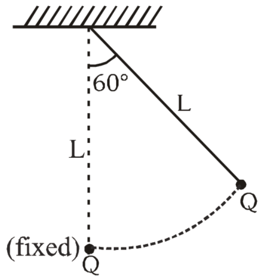
(1) 

(2) 

(3) 

(4) 

39. A charged metallic bob of mass m and charge Q is in equilibrium in presence of electric repulsion by Q (fixed) as shown in fig. Find Q = ?



(1) $(2\pi\epsilon_0 mg)^{1/2}L$ (2) $(4\pi\epsilon_0 mg)^{1/2}L$

(3) $(8\pi\epsilon_0 mg)^{1/2}L$ (4) None

40. Eight equal charges each +Q are kept at the corners of a cube. Net electric field at the centre will be $\left(k=\dfrac{1}{4\pi\varepsilon_0}\right)$

(1) $\dfrac{kQ}{r^2}$ (2) $\dfrac{8kQ}{r^2}$ (3) $\dfrac{2kQ}{r^2}$ (4) Zero

41. Two charges each equal to q are kept at x = – a and x = a on the X axis. A particle of mass m and charge $q_0=\dfrac{q}{2}$ is placed at the origin if charge q_0 is given a small displacement (y << a) along the y–axis, the net force acting on the particle is proportional to :-

(1) $\dfrac{1}{y}$ (2) $-\dfrac{1}{y}$ (3) y (4) – y

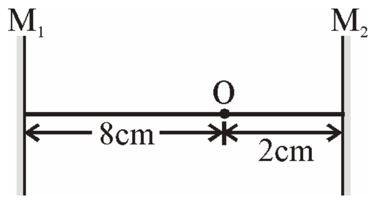
42. If $q_1 + q_2 = q$, then the value of the ratio $\dfrac{q_1}{q}$, for which the force between q_1 and q_2 is maximum is:

(1) 0.25 (2) 0.75 (3) 1 (4) 0.5

43. An object is at a distance of 0.5 m in front of a plane mirror. Distance between the object and image is :-

(1) 0.5 m (2) 1 m (3) 0.25 m (4) 1.5 m

44. Figure below shows two plane mirrors and an object O placed between them. What will be distance of the first three images from the mirror M_2 ?



(1) 2 cm, 8 cm, 14 cm (2) 2 cm, 12 cm, 18 cm
(3) 2 cm, 18 cm, 22 cm (4) 2 cm, 24 cm, 38 cm

45. The refractive indices of glass and water w.r.t air are 3/2 and 4/3 respectively. The refractive index of glass w.r.t water will be

(1) $\frac{8}{9}$ (2) $\frac{9}{8}$ (3) $\frac{7}{6}$ (4) None

46. A convex mirror has a radius of curvature 20 cm. If a point source is placed 14cm away from the mirror, the image is formed

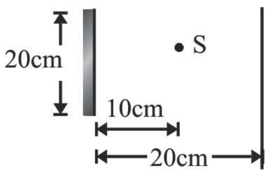
(1) $\frac{70}{12}$ cm behind the mirror

(2) $\frac{70}{12}$ cm in front the mirror

(3) $\frac{80}{12}$ cm behind the mirror

(4) 5 cm in front the mirror

47. A point source of light S is placed at a distance 10 cm infront of the centre of a mirror of width 20 cm suspended vertically on a wall. A man walks with a speed 10 cm/s infront of the mirror along a line parallel to the mirror at a distance 20 cm from it as shown in figure. Find the maximum time during which he can see the image of the source S in the mirror :-



(1) 6 s (2) 3 s (3) 2 s (4) 1 s

48. A fish in water sees an object which is 24 cm above the surface of water. The height of the object above the surface of water that will appear to the fish is ($\mu = 4/3$)

(1) 24 cm

(2) 32 cm

(3) 18 cm

(4) 48 cm

49. A beam of light from a source is incident normally on a plane mirror fixed at a distance 2m from the source. The beam is reflected back as a spot on a scale placed just above the source. When the mirror is rotated through a small angle 1.8°, the spot of the light is found to move through a distance d on the scale. The distance d is :-

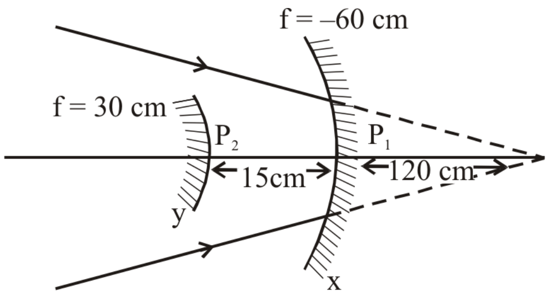
(1) 0.1831 m

(2) 0.1256 m

(3) 3.14 m

(4) None of these

50. The image formed after two reflection (first at x then at y) will be :-



(1) Erect and enlarged 2 times

(2) Inverted and enlarged 2 times

(3) Erect and diminished 2 times

(4) Inverted and diminished 4 times

SUBJECT : CHEMISTRY

Topic : PERIODIC TABLE (COMPLETE), CHEMICAL KINETICS

SECTION-A

Attempt All 35 questions

51. First IE of 5d series elements are higher than those of 3d and 4d series elements. This is due to :

(1) Bigger size of atoms of 5d-series elements than 3d-series elements.

(2) Greater effective nuclear charge is experienced by valence electrons because of the weak shielding by 4f-electrons in 5d series.

(3) (1) and (2) both.

(4) None of these.

52. Out of N, O, Ne, Na, Ne^+ select the species having maximum & minimum I.P. respectively :-

(1) Ne, Ne^+ (2) Ne^+, Na

(3) Ne, Na (4) Ne, N

53. Select the correct order of size for :
Cl^-, O^{-2}, Na^+, Br^-, Li^+

(1) $Br^- > O^{-2} > Cl^- > Na^+ > Li^+$

(2) $Cl^- > Br^- > O^{-2} > Na^+ > Li^+$

(3) $Br^- > Na^+ > Li^+ > O^{-2} > Na^+$

(4) $Br^- > Cl^- > O^{-2} > Na^+ > Li^+$

54. The element with strongest electropositive nature is :

(1) Cu (2) Cs (3) Cr (4) Ba

55. Among the following configurations, the element which has the highest electron affinity is :-

(1) [Ne] $3s^23p^2$ (2) [Ne] $3s^23p^5$

(3) [Ne] $3s^23p^4$ (4) [Ne] $3s^23p^1$

56. Match the following

	Order		Property
(a)	Li < Be < B	(i)	Electron affinity
(b)	Be < B < Li	(ii)	Atomic radius
(c)	Li < B < Be	(iii)	Ionisation energy
(d)	Li > Be > B	(iv)	Z_{eff}

(1) a – (iv); b – (i); c – (ii); d – (iii)

(2) a – (i); b – (iv); c – (ii); d – (iii)

(3) a – (iv); b – (i); c – (iii); d – (ii)

(4) a – (iv); b – (iii); c – (i); d – (ii)

57. The last element of the p-block in the 6th period represented by the outermost electronic configuration

(1) $7s^2$ 7p6 (2) $5f^4\ 6d^{10}\ 7s^2\ 7p^\circ$

(3) $4f^{14}\ 5d^{10}\ 6s^2\ 6p^6$ (4) $4f^4\ 5d^{10}6s^2\ 6p^4$

58. Magnetic moment of X^{3+} ion of 3d series is $\sqrt{35}$ BM. What is atomic number of X^{3+} ?

(1) 25 (2) 26 (3) 27 (4) 28

59. The pair with minimum difference in electronegativity is :-

(1) F, Cl (2) C,H (3) P, H (4) Na, Cs

60. Select equation having exothermic step
(I) $S^+_{(g)} \rightarrow S_{(g)}$ (II) $N^+_{(g)} \rightarrow N\ (g)$
(III) $N_{(g)} \rightarrow N^-_{(g)}$ (IV) $Al^{+2}_{(g)} \rightarrow Al^{+3}_{(g)}$
Choose the correct code

(1) II (2) I,II

(3) III and IV (4) II and III

61. The correct order of Radius is -

(1) $Li^+ < Be^{+2}$ (2) $Mn > Sc$

(3) $Li^+ \simeq Mg^{+2}$ (4) $Na < B$

62. Choose the correct matching

	Column-I	Column-II
(1)	$[Xe]\ 4f^{14}5d^{7}6s^{2}$	Une
(2)	$5s^{2}5p^{2}$	Ba
(3)	$[Kr]\ 4d^{10}5s^{2}$	Cd
(4)	$[Ne]\ 3d^{5}4s^{1}$	Cu

63. Select the incorrect statement

(a) P-orbital can accommodate 6es$^-$

(b) $Na \xrightarrow{IE_1} Na^{+} \xrightarrow{IE_2} Na^{+2} \xrightarrow{IE_3} Na^{+3}$;

The correct order of Ionisation Energy is $IE_3 > IE_1 > IE_2$

(c) Gd is Inner transition element.

(d) B is non metal.

(1) Only a (2) a and b

(3) b and c (4) a,b and d

64. Which of the following ion has maximum Electron affinity -

(1) O^+ (2) S^+ (3) Cl^+ (4) F^+

65. How many total elements from the following are not transition elements.

Zr, Co, Ge, Cd, Hg, Au, Cs

(1) 5 (2) 4 (3) 6 (4) 3

66. Which of the following ion has the highest value of Ionic Radius -

(1) O^{2-} (2) B^{3+} (3) Li^+ (4) Mg^{+2}

67. Which of following atom has maximum difference of Ionisation energy-

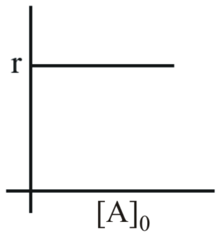
(1) B, Tl (2) Al, Ga (3) B, In (4) Ga, In

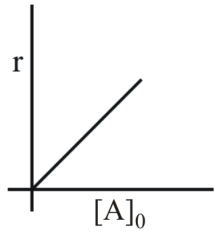
68. Select the correct order of Atom\Ionic Radius is -

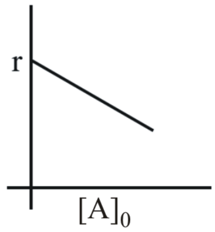
(1) $Zr^{+4} \simeq Hf^{+4}$ (2) $Eu^{+3} > Gd^{+3}$

(3) $Eu > Gd$ (4) All of these

69. For Ist order reaction, correct graph between rate and concentration of reactant will be :

(1) 

(2) 

(3) 

(4) None

70. ***Assertion :*** For a first order reaction $t_{1/2}$ is independent of the inital concentration of reactants.

Reason : For a first order reaction $t_{1/2}$ is twice the $t_{3/4}$.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

71. At 300 K the half-life of a sample of a gaseous compound initially at 1 atm is 100 s. When the pressure is 0.5 atm the half-life is 50 s. The order of reaction of compound is :-

(1) 0 (2) 1 (3) 2 (4) 3

72. For reaction A $\rightarrow$ B rate constant is 1.2×10^{-2} Ms^{-1}. What is concentration of B after 10 minute. If we start with 10 M of A :-

(1) 2.8 M (2) 7.2 M (3) 8.2 M (4) 2.7 M

73. If $\frac{d[NH_3]}{dt} = 34$ g hr^{-1} for the reaction

$N_2 + 3H_2 \rightarrow 2NH_3$

Then $-\frac{d[H_2]}{dt}$ is ....g h^{-1}.

(1) 6 (2) 4 (3) 51 (4) 12

74. For a I^{st} order reaction what will be the time required to reach concentration from 1 M to 0.25 M if $t_{1/2}$ is 138 s :-

(1) 276 s (2) 138 s

(3) 69 s (4) 27.6 s

75. For a reaction A + B $\rightarrow$ C + 2D, experimental results were collected for three trials and the data obtained are given below

Trial	[A], M	[B], M	Initial Rate, M s^{-1}
(1)	0.40	0.20	55×10^{-4}
(2)	0.80	0.20	5.5×10^{-3}
(3)	0.80	0.40	2.2×10^{-2}

The correct rate law of the reaction is :

(1) Rate = $k[A]^0 [B]^2$ (2) Rate = $k[A][B]^2$

(3) Rate = $k[A][B]$ (4) Rate = $k[A][B]^0$

76. In a reaction 2A $\rightarrow$ Products, the Concentration of A decreases from 0.8 mol L^{-1} to 0.3 mol L^{-1} in 5 minutes then what is the average rate during this interval ?

(1) 2.5×10^{-2} $M.min^{-1}$ (2) 10^{-1} $M.min^{-1}$

(3) 1.25×10^{-2} $M.min^{-1}$ (4) 5×10^{-2} $M.min^{-1}$

77. **Statement-I :** The rate of a first order reaction decreases with passage of time as the concentration of reactant decreases.

Statement-II : Value of rate constant decreases with decrease in concentration of reactant.

(1) Statement-I and Statement-II both are true.

(2) Statement-I is false and Statement-II is true.

(3) Statement-I is true and Statement-II is false.

(4) Statement-I and Statement-II both are false.

78. The initial concentration of N_2O_5 in following first order reaction $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$ was 2.48×10^{-2} mol L^{-1} at 318K. The concentration of N_2O_5 after 120 minutes was 0.40×10^{-2} mol L^{-1}. Find the rate constant of reaction at 318 K.

(1) 0.0304 min^{-1} (2) 0.0152 min^{-1}

(3) 0.0076 min^{-1} (4) 0.0608 min^{-1}

79. Which of the following statement is incorrect –

(1) In complex reaction molecularity of the slowest step is same as the order of the overall reaction if intermediate is not present.

(2) The decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at low pressure.

(3) The fraction of molecules having energy equal to or lower than threshold value are known as activated molecules.

(4) Both 2 & 3 are incorrect.

80. The activation energy for the reaction.

$2AB(g) \rightarrow A_2(g) + B_2(g)$ is 4K cal mol^{-1} at 500K. Find fraction of molecules of reactants having energy equal to or greater than activation energy ?

(1) 1.8×10^{-2} (2) 2.8×10^{-2}

(3) 1.8×10^{-3} (4) 3.8×10^{-2}

81. The rate of a reaction quadruples when the temperature changes from 290 K to 310 K.
Find the energy of activation of the reaction assuming that it does not change with temperature ?

(1) $\dfrac{2.303R \times 310 \times 290}{20}$

(2) $\dfrac{0.693R \times 310 \times 290}{20}$

(3) $\dfrac{1.38R \times 310 \times 290}{20}$

(4) $\dfrac{20}{1.38R \times 310 \times 290}$

82. A first order reaction takes 60 min. for 99.9% decomposition then half life is :

(1) 60 min

(2) 3 min

(3) 2 min

(4) 6 min

83. Match the terms of column-I with column-II

	Column-I		Column-II
A	Hydrolysis of ester in acidic medium	p	Zero order reaction
B	Radioactive decay	q	Pseudo first order reaction
C	Thermal decomposition of HI on gold surface	r	First order reaction

(1) A → p, B → q, C → r

(2) A → q, B → r, C → p

(3) A → r, B → q, C → p

(4) A → q, B → p, C → r

84. The half life for radioactive decay of carbon-14 is 693 years. An archeological artifact containing wood had only 90% of the C^{14} found in a living tree estimate the age of the sample.

(1) 1151.5 years

(2) 2303 years

(3) 4606 years

(4) 6930 years

85. The decomposition of ammonia on platinum surface is zero order reaction what is the rate of production of H_2.
If $K = 5 \times 10^{-4}$ mol L^{-1} S^{-1} ?
$2NH_3 \rightarrow N_2 + 3H_2$

(1) 3×10^{-4} mol L S^{-1}

(2) 5×10^{-4} mol L S^{-1}

(3) 1.5×10^{-3} mol^{-1} L S^{-1}

(4) 1.5×10^{-3} mol L^{-1} S^{-1}

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

86. The incorrect option or statement is :-

(1) The equation of 2^{nd} IP is $X(g) \rightarrow X^{+2} + 2e^-$

(2) 2^{nd} IP of an element is lower than its 3^{rd} IP

(3) $Cl^- > F^-$ (I.E)

(4) Ionisation enthalpies are always positive for a neutral atom.

87. The correct match of contents in Column-I with those in Column-II is :

	Column-I		Column-II
(A)	He	(i)	High electron gain enthalpy
(B)	Cl	(ii)	Electropositive element
(C)	Ca	(iii)	IA group
(D)	Li	(iv)	Highest ionisation energy

(1) A-(iii), B-(i), C-(ii), D-(iv)

(2) A-(iv), B-(iii),C-(ii), D-(i)

(3) A-(ii), B-(iv), C-(i), D-(iii)

(4) A-(iv), B-(i),C-(ii), D-(iii)

88. **Assertion (A):** The ionic radii of O^{2-} and Mg^{2+} are same.

Reason (R): Both O^{2-} and Mg^{2+} are isoelectronic species.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

89. Which of the following process is an endothermic process

(1) $O(g) + e^- \rightarrow O^-(g)$

(2) $Cl(g) + e^- \rightarrow Cl^-(g)$

(3) $N(g) + e^- \rightarrow N^-(g)$

(4) $S(g) + e^- \rightarrow S^-(g)$

90. Read the following statements -

(a) II^{nd} I.E. will be always greater than I^{st} I.E.

(b) Size order $Mg > A\ell > Mg^{+2} > Al^{+3}$

(c) Group 17 elements have very less negative electron gain Enthalpy.

(d) In two compound OF_2 and Na_2O the correct order of Electronegativity is $F > O > Na$

How many total statements are correct -

(1) 2 (2) 3 (3) 4 (4) 1

91. Select the incorrect order of Electron gain enthalpy (EA)

(1) $P < S$ (2) $N < O$ (3) $N^+ < N$ (4) $O^- < O$

92. Which of the following is the correct order according to their given property -

(1) $O^{-2} > Br^- > Se^{-2}$ Order of Radius

(2) $C < N > O < F$ Order of Ionisation Energy

(3) $Li < B$ Order of Electron affinity.

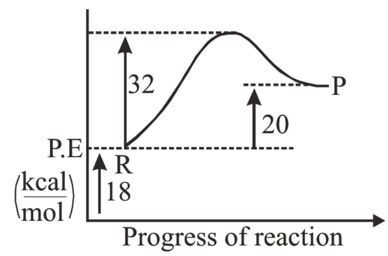
(4) $NaF > CsF$ Ionic Nature.

93. 50% of first order reaction completed in 16 min then 75% of reaction completed in

(1) 30 min (2) 32 min

(3) 64 min (4) 16 min

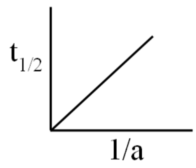
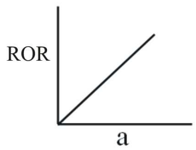
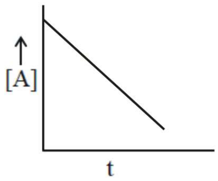
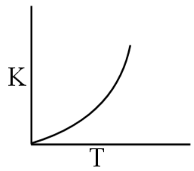
94. What is the threshold energy of reaction ($R \rightarrow P$) in represented diagram? (all energy in kcal/mol) :-



(1) 12 kcal/mol (2) 14 kcal/mol

(3) 50 kcal/mol (4) 70 kcal/mol

95. Match the following :-

Column-I		Column-II	
(i)		(P)	Zero order
(ii)		(Q)	First order
(iii)		(R)	Second order
(iv)		(S)	$K = Ae^{-\frac{Ea}{RT}}$

(1) (i)→P, (ii)→Q, (iii)→R, (iv)→S
(2) (i)→S, (ii)→R, (iii)→Q, (iv)→P
(3) (i)→R, (ii)→Q, (iii)→P, (iv)→S
(4) (i)→Q, (ii)→P, (iii)→S, (iv)→R

96. A gaseous reaction $A_{2(g)} \longrightarrow B_{(g)} + \frac{1}{2}C(g)$
Shows increase in pressure from 100 mm of Hg to 120 mm of Hg in 5 minute. What will be rate of dissappearance of A_2 (in mm of Hg/min) :-

(1) 8 (2) 4 (3) 40 (4) 10

97. The hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below :-
(1) $A_2 \leftrightharpoons A + A$ ....(fast)
(2) $A + B_2 \rightarrow AB + B$.....(slow)
(3) $A+B \rightarrow AB$-----(fast)
The order of the reaction is :-

(1) 1.5 (2) 2 (3) Zero (4) 1

98. The factors affecting the rate constant of reaction are –
I. Temperature
II. Pressure
III. Concentration of reactant or product
IV. Catalyst
Choose the correct option from the alternative given below.

(1) I, II and III
(2) I, III and IV
(3) I, IV
(4) All of these

99. In a zero order reaction if the temperature is increased from 10°C to 40°C. The rate of reaction will become :–

(1) 512 times (2) 8 times
(3) 256 times (4) 64 times

100. Consider a first order gas phase decomposition reaction given below :
$A(g) \rightarrow B(g) + C(g)$
The initial pressure of the system before decomposition of A was P_i After lapse of time 't' total pressure is P_t.
The rate constant of reaction is :–

(1) $K = \frac{2.303}{t}\log\frac{P_i}{P_i - x}$

(2) $K = \frac{2.303}{t}\log\frac{P_i}{2P_i - P_t}$

(3) $K = \frac{2.303}{t}\log\frac{P_i}{2P_i + P_t}$

(4) $K = \frac{2.303}{t}\log\frac{P_i}{P_i + x}$

SUBJECT : BIOLOGY-I

Topic : HUMAN REPRODUCTIVE SYSTEM, LIFE CYCLE OF ANGIOSPERMS

SECTION-A

Attempt All 35 questions

101. In the given list how many structures are included in male reproductive system ?
Testes, Accessory ducts, Accessory glands, External genitalia.

(1) 3 (2) 4
(3) 2 (4) 1

102. Which of the following statement is false ?

(1) Humans are sexually reproducing and viviparous.
(2) Implantation is followed by formation of blastocyst.
(3) Sperm formation continues even in old man.
(4) Gestation is followed by parturition.

103. Secretions of which of the following glands helps in lubrication of penis ?

(1) Prostate gland (2) Bulbourethral gland
(3) Both (1) and (2) (4) Seminal vesicle

104. Which of the following is not a part of female accessory ducts ?

(1) Vagina (2) Mons pubis
(3) Oviduct (4) Uterus

105. The last part of oviduct which has a narrow lumen and joins the uterus is ?

(1) Ampulla (2) Infundibulum
(3) Isthmus (4) Fimbriae

106. Which of the following pair is not matched correctly ?

(1) Length of fallopian tube ⇒ 10-12 cm
(2) Number of testicular lobules in a male ⇒ 500
(3) Width of testis ⇒ 4-5 cm
(4) Length of ovary ⇒ 2-4 cm

107. Which layer of uterus undergoes cyclical changes during menstrual cycle ?

(1) Perimetrium
(2) Tunica vasculosa
(3) Myometrium
(4) Endometrium

108. Tiny finger like structure which lies at the upper junction of two labia minora is ?

(1) Mons pubis (2) Hymen
(3) Labia majora (4) Clitoris

109. **Statement-I :** Formation of ovum ceases in women around the age of fifty years.
Statement-II : Each testicular lobule contains 1-3 highly coiled seminiferous tubules in which sperm are produced.
Choose the correct answer from the given options :-

(1) Both statement-I and II are correct
(2) Both statemen-I and II are incorrect
(3) Statement-I is correct but statement II is incorrect
(4) Statement-I is incorrect but statement II is correct

110. Nurse cells are also termed as ?

(1) Sertoli cells (2) Leydig cells

(3) Interstitial cells (4) Spermatogonia

111. In the given list how many structures are present inside the testes ?

Immunologically competent cell, Leydig cells, Sertoli cells, Spermatogonia.

(1) 2 (2) 4 (3) 1 (4) 3

112. Which of the following hormone is released in case of excess spermatogenesis ?

(1) ABP (2) Testosterone

(3) FSH (4) Inhibin

113. Which statement is false for sperm ?

(1) A plasma membrane envelops only the head of sperm.

(2) The sperm head containing a haploid elongated nucleus.

(3) The middle piece contains numerous mitochondria.

(4) It is a microscopic structure

114. Which of the following is characterised by a fluid filled cavity called antrum ?

(1) Primordial follicle (2) Secondary follicle

(3) Tertiary follicle (4) Primary follicle

115. Which of the following is not another name of luteal phase of menstrual cycle ?

(1) Secretory phase

(2) Proliferative phase

(3) Progesteronic

(4) Postovulatory phase

116. Which of the following pair is not matched correctly ?

(1) Primary follicles in a female at puberty ⇒ 1,20,000 - 1,60,000

(2) First menstruation at puberty ⇒ Menarche

(3) Meiotic first division of oogenesis ⇒ Equal division

(4) Arrest of meiosis-I during oogenesis ⇒ Prophase-I

117. In a female if there is no entry of sperm then how many structures in the given list will not be formed ?

1^{st} polar body, 2^{nd} polar body, Secondary oocyte, Primary oocyte.

(1) 2 (2) 1 (3) 4 (4) 3

118. Read the following statement (A-D) :-

(A) Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.

(B) Secondary oocyte retains the bulk of the nutrient rich cytoplasm of the primary oocyte.

(C) In testis, the immature germ cells (spermatogonia) produce sperms by spermatogenesis that begins at puberty.

(D) Primary oocyte is surrounded by a layer of granulosa cells called the primary follicles.

Which of the above statements are true :-

(1) A, B, C and D (2) B only

(3) B,C and D only (4) A and C only

119. In oogenesis, the 2^{nd} polar body is formed inside which structure ?

(1) Ovary (2) Uterus

(3) Peritoneal cavity (4) Fallopian tube

120. Fusion of pronuclei of male and female gametes is ?

(1) Amphimixis (2) Plasmogamy
(3) Karyogamy (4) Ovulation

121. Which of the following hormone leads to increase in antral fluid in the ovarian follicles ?

(1) ABP (2) LH
(3) Progesterone (4) Inhibin

122. Which of the following hormone increases the secretory nature of endometrium ?

(1) FSH (2) LH
(3) Progesterone (4) Estrogen

123. Which of the following statement is true ?

(1) Each seminiferous tubule is lined on the out side by germinal epithelium.

(2) The hymen can never be torn by a sudden fall or jolt or insertion of vaginal tampon.

(3) In humans, maturation phase is the longest phase of oogenesis.

(4) On the 23^{rd} day of menstrual cycle the thickness of uterine wall is minimum

124. **Assertion (A) :** Breast feeding during the initial period of infant growth is recommended by doctors.
Reason (R) : Colostrum contains antibodies (mainly IgA) absolutely essential to develop resistance for the new born babies.
Choose the correct answer from the given options :-

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

125. **Assertion (A) :** The presence of hymen is not a reliable indicator of virginity.
Reason (R) : In some women hymen persists even after coitus.
Choose the correct answer from the given options :-

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) correct and (R) is the correct explanation of (A).

126. The regions outside the seminiferous tubules called interstitial spaces contains :-

(1) Small blood vessels

(2) Interstitial cells

(3) Immunologically competent cells

(4) All of the above

127. Select the correct match from following table.

(A)	ICSH	(i)	Released by hypothalamus
(B)	FSH	(ii)	Stimulate leydig cells
(C)	ABP	(iii)	Stimulate Sertoli cells
(D)	GnRH	(iv)	Released by sertoli cells

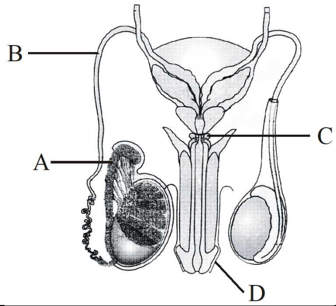
(1) A – ii ; B – iii ; C – iv ; D – i

(2) A – iii ; B – ii ; C – i ; D – iv

(3) A – i ; B – ii ; C – iii ; D – iv

(4) A – iv ; B – iii ; C – ii ; D – i

128. It is a digrammatic view of male reproductive system. Choose the correct option about A, B, C and D.



	A	B	C	D
(1)	Vasa efferentia	Vas deferens	Prostate	Foreskin
(2)	Epididymis	Vasa efferentia	Seminal Vesicle	Glans penis
(3)	Rete testis	Ureter	Bulbo-Urethral gland	Glans penis
(4)	Epididymis	Vas deferens	Bulbo-Urethral gland	Foreskin

129. 50 secondary oocytes in female and 50 secondary spermatocytes in male give rise to :-

(1) 100 ova and 100 sperms

(2) 200 ova and 50 sperms

(3) 100 ova and 200 sperms

(4) 50 ova and 100 sperms

130. Capacitation refers to changes in the :-

(1) Ovum before fertilization.

(2) Ovum after fertilization.

(3) Sperm after fertilization.

(4) Sperm before fertilization.

131. The correct sequence of spermatogenetic stage leading to the formation of sperm in a mature human testis is :-

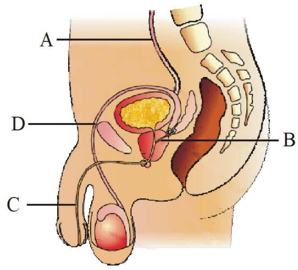
(1) Spematogonia - Spermatids - Spermatocytes - Sperms

(2) Spermatocyte - Spematogonia - Spermatids - Sperms

(3) Spematogonia -Spermatocytes - Spermatids - Sperms

(4) Spermatids - Spermatocytes - Spematogonia - Sperms

132. It is a diagrammatic sectional view of male reproductive system, In which identify common path for the semen and urine :



(1) A (2) B (3) C (4) D

133. Seminal plasma in human is rich in :

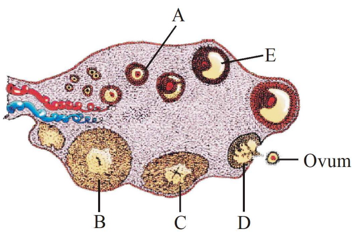
(1) Glucose, fructose, Ribose

(2) Fructose, Calcium, Certain enzymes

(3) Ribose, Testosterone, Mucus

(4) Testosterone, ABP and Sperms

134. Observe the following diagram and answer :



Which structure secrete progesterone :-

(1) A and E (2) E and D

(3) B and E (4) B and C

135. Which statement is incorrect ?

(1) The stroma of ovary is divided into two zones, cortex and medulla.

(2) In the later phase of pregnancy, a hormone called relaxin is also secreted by the ovary.

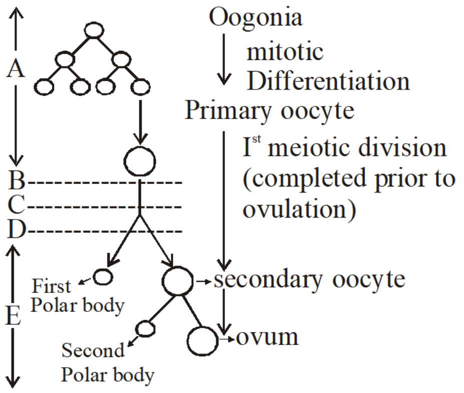
(3) Corona radiata is a primary egg membrane.

(4) Hyaluronic acid is found between the cells of corona radiata.

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

136. The following refers to schematic representation of oogenesis. Identify A to E correctly



(1) (A) Fetal life (B) Birth (C) Puberty (D) Adult reproductive life (E) child hood

(2) (A) Fetal life (B) Birth (C) Childhood (D) Puberty (E) Adult reproductive life

(3) (A) Adult reproductive life (B) Birth (C) puberty (D) child hood (E) Fetal life

(4) (A) Birth (B) child hood (C) Fetal life (D) puberty (E) adult reproductive life

137. Which statement is not correct for both testes and ovary :

(1) Both are primary sex organs.

(2) Both are developed from mesoderm.

(3) Both are developed in abdominal cavity in Intra uterine life.

(4) At the time of birth both descend down into extra abdominal space.

138. The _____ are the first haploid cells during the process of spermatogenesis.

(1) Spermatogonia

(2) Primary spermatocyte

(3) Secondary spermatocyte

(4) Spermatids

139. Choose the correct option for filling up the blanks:-

The human male ejaculates about ____ million sperms during a coitus of which, for normal fertility, at least _____ percent sperms must have normal shape and size and least _____ percent of then must show vigorous motility.

(1) 100–200, 40, 60 (2) 200–300, 60, 40

(3) 300–400, 50, 30 (4) 500, 70, 70

140. Choose the correct order for the path of sperm from the testes to outside the body.

(1) vas deferens - epididymis - ejaculatory duct - penis

(2) epididymis - vas deferens - ejaculatory duct - penis

(3) ejaculatory duct - vas deferens - epididymis - penis

(4) penis - ejaculatory duct - epididymis - vas deferens

141. Arrange following events of female reproductive cycle in (human) is sequence :-

(A) Secretion of FSH

(B) Growth of corpus luteum

(C) Growth of follicles

(D) Ovulation

(E) Sudden increase in level of LH

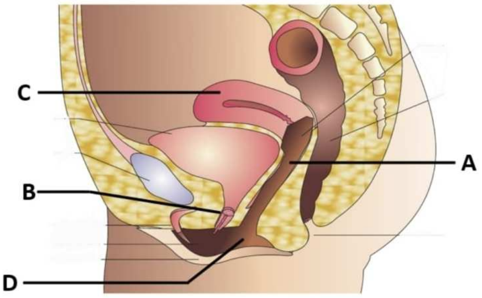
(1) $C \rightarrow A \rightarrow D \rightarrow B \rightarrow E$

(2) $A \rightarrow C \rightarrow E \rightarrow D \rightarrow B$

(3) $A \rightarrow D \rightarrow C \rightarrow E \rightarrow B$

(4) $B \rightarrow A \rightarrow C \rightarrow D \rightarrow E$

142. Given figure is a diagrammatic sectional view of female pelvis showing reproductive system identify A,B,C and D :-



	A	B	C	D
(1)	Vagina	Urethra	Uterus	Vaginal orifice
(2)	Vaginal orifice	Cervix	Urinary bladder	Labia majora
(3)	Cervix	Urethra	Uterus	Vagina
(4)	Urethra	Vagina	Urinary bladder	Urethral orifice

143. Vasa efferentia are the ductules those connect :-

(1) testicular lobules to rete testes

(2) rete testes to epididymis

(3) vas deferens to epididymis

(4) epididymis to urethra

144. Correct sequence for milk ejection is :

(1) Mammary tubule – Mammary duct – Lactiferous duct – Ampulla – Nipple

(2) Mammary duct – Mammary tubule – Ampulla – Lactiferous duct – Nipple

(3) Mammary tubule – Mammary duct – Ampulla – Lactiferous duct – Nipple

(4) Lactiferous duct – Mammary tubule – Mammary duct – Ampulla – Nipple

145. Choose the **incorrect** statement :-

(1) Oogonia are formed and added after birth

(2) Oogenesis is initiated during the embryonic development

(3) No more mother gamete cells are formed in females after birth

(4) 60000–80000 Primary follicle are present in each ovary at puberty

146. How many of the following structures in the given list contain 46 chromosomes ?

Spermatozoa, secondary oocyte, spermatogonia, secondary spermatocyte, primary spermatocyte, spermatid, polar body, primary oocyte, oogonia

(1) Six (2) Four

(3) Five (4) Seven

147. Read the following statements carefully.

The ....(A)...... present on the inside wall of ....(B)....... multiply by .....(C)...... division. A ......(D)....... completes the ....(E)....... division leading to formation of two equal, haploid cells called ....(F)....... .

Which one of the following options gives the correct filling of A, B, C, D, E and F?

	(1)	(2)	(3)	(4)
(A)	Spermatogonia	Primary Spermatocyte	Spermatogonia	Primary Spermatocyte
(B)	Seminiferous tubule	Sertoli Cell	Sertoli Cell	Seminiferous tubule
(C)	Mitotic	Mitotic	First meiotic	First meiotic
(D)	Primary Spermatocyte	Secondary Spermatocyte	Secondary Spermatocyte	Spermatid
(E)	First meiotic	First meiotic	Second meiotic	Second meiotic
(F)	Secondary Spermatocyte	Spermatid	Spermatid	Sperm

148. Match the following and choose the correct answer :-

(A)	Spermatogensis	(i)	Vagina
(B)	Capacitation	(ii)	Ovary
(C)	Folliculogenesis	(iii)	Testis
(D)	Fertilisation	(iv)	Fallopian tube

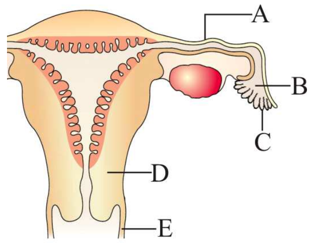
(1) A-(iv), B-(i), C-(iii), D-(ii)

(2) A-(iii), B-(ii), C-(iv), D-(i)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(iii), B-(iv), C-(ii), D-(i)

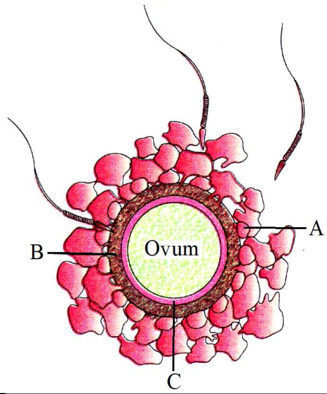
149.



See the diagram and Identify A,B,C,D,E :-

	A	B	C	D	E
(1)	Infundibulum	Ovary	Fimbriae	Cervix	Vagina
(2)	Fallopian tube	Infundibulum	Cervix	Vagina	Isthmus
(3)	Isthmus	Infundibulum	Fimbriae	Cervix	Vagina
(4)	Cervix	Ovary	Vagina	Fimbriae	Isthmus

150. It is a digramatic presenation of ovum which is surrounded by few sperms, choose the correct option about A, B, and C :-



	A	B	C
(1)	Vitelline membrane	Zona Pellucida	Cells of the corona radiata
(2)	Zona Pellucida	Vitelline membrane	Cells of the corona radiata
(3)	Plasma membrane	Vitelline membrane	Perivitelline space
(4)	Cells of the corona radiata	Zona Pellucida	Perivitelline space

SUBJECT : BIOLOGY-II

Topic : HUMAN REPRODUCTIVE SYSTEM, LIFE CYCLE OF ANGIOSPERMS

SECTION-A

Attempt All 35 questions

151. Polycarpic plants are :-

(1) Annual (2) Biennial

(3) Perennial (4) Fruiting once

152. Pollen grains :-

(1) Are generally tetragonal

(2) Measuring 25-50 micrometers in diameter

(3) Has prominent three layered wall

(4) Formed inside megasporangium

153. Functional megaspore develops into polygonum type embryosac by

(1) One meiosis and two succcessive mitosis

(2) Two meiosis and two successive mitosis

(3) One meiosis and three successive mitosis

(4) Three successive mitosis

154. Select the **incorrect** pair :-

(1) Embryo sac – Microgametophyte

(2) Pollen grain – Microspore

(3) Stamen – Microsporophyll

(4) Ovule – Megasporangium

155. One of the most resistant organic material present in the exine of pollen grain is :-

(1) Pectocellulose (2) Sporopollenin

(3) Suberin (4) Cellulose

156. Pollen tablets are used as ______

(1) As antibiotic

(2) As antiseptic

(3) As food supplement

(4) Anti allergen

157. In flowering plants, two male gametes are formed by mitotic division in

(1) Generative cell (2) Central cell

(3) Synergid cell (4) Tube cell

158. Which of the following is not a feature of typical angiospermic anther ?

(1) It consist of four microsporangia

(2) It is situated on distal end of filament

(3) It is site for microspore formation

(4) At maturity, when it dehisce, then it has two pollen sacs in each lobe

159. Tapetum cells of anther are usually :-

(1) Diploid, uninucleate

(2) Haploid, binucleate

(3) Haploid, tetranucleate

(4) Polyploid, multinucleate

160. Typical anther of flowering plant is :

(1) Dithecous and tetrasporangiate

(2) Dithecous and bisporangiate

(3) Monothecous and bisporangiate

(4) Monothecous and monosporangiate

161. Residual, persistant nucellus is :-

(1) Perisperm (2) Endosperm

(3) Embryo sac (4) Aril

162. The egg apparatus in an embryo sac consists of

(1) Antipodal + egg cell

(2) Synergids + egg cell

(3) Synergids + antipodals + egg cell

(4) Synergids + polar nuclei + egg cell

163. How many meiotic divisions are required to form 200 mature male gametophytes in angiosperm from microspore mother cell ?

(1) 50 (2) 25 (3) 100 (4) 200

164. First cell of female gametophyte is :-

(1) Functional megaspore

(2) Microspore mother cell

(3) Megaspore mother cell

(4) Egg cell

165. Given below are two statements :-

Statement-I : Geitonogamy is functionally cross-pollination involving a pollinating agent.

Statement-II : Geitonogamy is genetically similar to autogamy since pollen grains come from the same plant.

Choose correct answer from the options given below :-

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are correct.

(4) Statement-I is incorrect but statement-II is correct.

166. In which one of the following, both autogamy and geitonogamy are prevented ?

(1) Wheat (2) Papaya

(3) Castor (4) Maize

167. What is true about cleistogamy ?

(1) It occurs in unisexual flowers

(2) It produces assured seed-set even in the absence of pollinators

(3) It leads to introduction of new useful characters

(4) It is a method of geitonogamy

168. Which one of the following statements is not true ?

(1) Micropyle facilitates entry of pollen tube into seed, during germination.

(2) Pollen grains are rich in nutrients, and they are used in the form of tablets and syrups

(3) Pollen grains of some plants cause severe allergies and bronchial afflictions in some people

(4) The flowers pollinated by flies and bats secrete foul odour to attract them

169. Maize and grasses show :-

(1) Entomophily (2) Hydrophily

(3) Chiropterophily (4) Anemophily

170. Which of the following is not an adaptation for anemophily ?

(1) Presence of well exposed stamens

(2) Often presence of single ovule in an ovary

(3) Sticky pollen grain

(4) Large and often feathery stigma

171. Features like large feathery stigma, non-sticky pollen grains and single ovule in each ovary are charecteristics of plants pollinated by :-

(1) Insect

(2) Water

(3) Air

(4) Birds

172. When on octaploid male plant is crossed with octaploid female plant then ploidy of endosperm and embryo would be, respectively

(1) 12n, 8n

(2) 8n, 12n

(3) 6n, 6n

(4) 8n, 8n

173. Which one of the following statement is not true?

(1) In the grass family cotyledon is called scutellum.

(2) Five nuclei and three male gametes participate in double fertilization

(3) Embryo develops at the micropylar end of embryo sac where the zygote is situated

(4) Orchid seeds are non-endospermic

174. Read the following statements carefully :-
(a) Pollen tube contains two male gametes
(b) One male gamete is motile and other is non motile
(c) Both male gametes are non-motile
(d) Pollen tube shows chemotropic movement
How many statements are correct about flowering plants ?

(1) 1 (2) 2 (3) 3 (4) 4

175. Pollen tube releases the two male gametes into the :

(1) One male gamete in cytoplasm of synergids and other in cytoplasm of central cell

(2) Both in cytoplasm of central cell

(3) Cytoplasm of antipodal cell

(4) Both in cytoplasm of synergids

176. Select correct for embryo development in flowering plants

(1) Develops at chalazal end of ovule

(2) Early stages of embryo development differs in monocots and dicots

(3) Most zygote divide only after certain amount of endosperm is formed

(4) Zygotes divide by meiosis division

177. Select the event from the following which is not referred as pollen pistil interaction :-

(1) Deposition of pollen on the stigma

(2) Entry of pollen tube into ovule

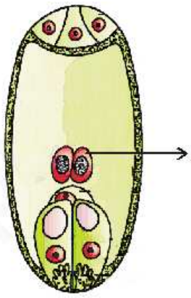
(3) Growth of pollen tube into the style

(4) Division of microspore mother cell in anther

178. How many of the following statements are true?
(A) Embryo development precedes endosperm development in angiosperm
(B) White kernel of coconut is free nuclear endosperm
(C) *Vallisneria* is a dioecious plants and flowers are unisexual.
(D) Formation of megaspore from megaspore mother cell is called megasporogenesis.

(1) Three (2) Four (3) Two (4) One

179. In the given figure, which of the following statements are true about the labelled region "A"



(a) Participate in syngamy
(b) Participate in triple fusion
(c) Participate in embryo formation
(d) Participate in endosperm formation

(1) a only
(2) b and d only
(3) a, b and d only
(4) a and b only

180. Identify the wrong statements regarding post fertilization developement :-

(1) The ovary wall develops into pericarp
(2) The ovary wall develops into seed coat
(3) The primary endosperm cell develops into endosperm
(4) The ovule develops into seed

181. **Assertion :-** Syngamy leads to formation of a specialised cell called zygote.

Reason :- One male gamete fertilise the egg cell in syngamy.

(1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

182. Arrange the following terms in the correct development sequence :-

(A) Male gametes
(B) Pollen grain
(C) Sporogenous tissue
(D) Pollen mother cell
(E) Microspore tetrad

(1) C, D, E, B, A
(2) B, C, D, A, E
(3) C, D, A, B, E
(4) B, D, C, A, B

183. During an experiment, vegetative nucleus was destroyed inside the pollen tube. Immediately afterwards it was observed that the growth of pollen tube was stopped and pollen tube could not reach up to ovule, because-

(1) Male gametes were not developed
(2) Growth of pollen tube is controlled by vegetative nucleus
(3) Vegetative nucleus secretes chemicals
(4) All of the above

184. Match the following

A.	Body of ovule fuses with funicle here	I.	Chalaza
B.	Basal part of the ovule	II.	Funicle
C.	Attached to placenta	III.	Nucellus
D.	Contains nutritive tissues	IV.	Hilum

(1) A – III, B – IV, C – I, D – II
(2) A – IV, B –I, C – II, D – III
(3) A –II, B – IV, C – I, D – III
(4) A – III, B – IV, C – I, D – II

185. Consider the following five statements (a-e) and select the option which includes all the correct ones only :-

(a) Vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus.

(b) Over 60 percent of angiosperms, pollengrains are shed at 3-celled stage.

(c) Pollen grains can be stored for years in liquid nitrogen (–196°C).

(d) Members of Rosaceae, Leguminosae and Solanaceae the pollen grains are viable for months.

(e) Pollen grains are poor in nutuients.

(1) a, b, c, d (2) a, c, d

(3) b, c, d (4) a, c, d, e

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. A flower has 10 microspore mother cells then how many pollen grains will be produced in that flower ?

(1) 50 (2) 10 (3) 20 (4) 40

187. In over 60 percent of Angiosperms, Pollen grains are shed at :

(1) 2-celled stage (2) 3-celled stage

(3) 4-celled stage (4) 1-celled stage

188. Aero allergens that came into India as a contaminant with imported wheat become ubiquitous in occurance :-

(1) *Chenopodium* (2) *Parthenium*

(3) *Amaranthus* (4) Sorghum

189. Filliform appratus is characteristic of :-

(1) Synergids (2) Egg

(3) Anther wall (4) Antipodals cells

190. In grass embryo, epicotyl has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure called :-

(1) Coleoptile

(2) Coleorhiza

(3) Scutellum

(4) Aleurone layer

191. One ovule is present in the ovary of:-

(1) Paddy

(2) Papaya

(3) Watermelon

(4) Orchids

192. Both chasmogamous and cleistogamous flowers are present in :-

(1) Sunflower (2) *Commelina*

(3) Pea (4) *Capsella*

193. Which of the following is not correct for double fertilization?

(1) Two types of fusion, syngamy and triple fusion take place

(2) The synergid cell after syngamy becomes the primary endosperm cell

(3) Double fertilization was discovered by Nawaschin

(4) It is of universal occurrence among angiosperms

194. Which of the following is a significance of triple fusion during double fertilisation in angiospermic plants?

(1) Essential for the formation of zygote

(2) Essential for the formation of endosperm

(3) Essential for the formation of embryo

(4) Essential for the formation of cotyledons in seed

195. The primary endosperm cell & zygote will be ________&______in a seed :-

(1) Embryo & Endosperm

(2) Embryo & Integuments

(3) Integuments & Embryo

(4) Endosperm & Embryo

196. Which one of the following is not a post-fertilisation event?

(1) Development of PEC into endosperm

(2) Development of ovary into fruit

(3) Maturation of ovule into seed

(4) Development of 7 celled-8 nucleate embryo sac

197. There are different developmental stages of embryo in dicot :-

(A) Globular embryo
(B) Heart shaped
(C) Zygote
(D) Proembryo
(E) Torpedo (mature embryo)
Arrange them in sequence

(1) C, D, A, B, E (2) B, A, C, D, E

(3) D, C, A, B, E (4) A, E, D, C, B

198. Match the following Column - I and II, select correct option indicating a correct match?

	Column-1 (Type of Pollination)		Column-II (Example)
(i)	Entomophily	(a)	*Zostera*
(ii)	Ephihydrophily	(b)	*Pinus*
(iii)	Hypohydrophily	(c)	*Amorphophallus*
(iv)	Anemophily	(d)	*Vallisneria*

(1) (i)-(b); (ii)-(d); (iii)-(c); (iv)-(a)

(2) (i)-(c); (ii)-(a); (iii)-(d); (iv)-(b)

(3) (i)-(c); (ii)-(d); (iii)-(a); (iv)-(b)

(4) (i)-(a); (ii)-(c); (iii)-(d); (iv)-(b)

199. Read the following statements and find out the incorrect statement.

(a) Plants use two abiotic (wind and water) and one biotic (animals) agents to achieve pollination.
(b) Majority of plants use abiotic agents for pollination.
(c) Only a small proportion of plants uses biotic agents.
(d) Pollination by water is common among abiotic pollinators.
(e) Pollination by wind is quite rare in flowering plants and is restricted to about 30 genera mostly monocotyledons.

(1) a, b, c and d (2) b, c, d and e

(3) a, c, d and e (4) b and d only

200. How many of the following structure in angiosperm plant have haploid set of chromosomes in cell ?

Microspore, generative cell, synergid, nucellus, integument, antipodal cell, archesporial cell, endosperm, zygote, vegetative cell of pollen

(1) Six (2) Eight

(3) Five (4) Three

Space for Rough Work

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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